

Alessandro Traspadini

Nationality: Italian **Date of birth:** 08/28/1997  **Email address:** alessandro.traspadini@unipd.it

EDUCATION AND TRAINING

Ph.D. Candidate in Information Engineering

University of Padova [10/01/2022 – 03/31/2026]

City: Padova | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies | **Level in EQF:** EQF level 8 | **Thesis:** Analysis, Design and Optimization of Non-Terrestrial Networks for 6G

Main research topics: Non-Terrestrial networks, 6G wireless networks, vehicular communications, mmWave communications.

Master's Degree in ICT for Internet and Multimedia - Telecommunications

University of Padova [09/30/2019 – 08/26/2021]

City: Padova | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies | **Final grade:** 110/110 cum laude | **Level in EQF:** EQF level 7 | **Thesis:** Scheduling for MU-MIMO using hybrid beamforming based on DPC

Master of Science in Communication Engineering (Double Degree)

National Taiwan University [09/01/2020 – 08/26/2021]

City: Taipei | **Country:** Taiwan | **Field(s) of study:** Information and Communication Technologies | **Final grade:** 4 (over 4) | **Level in EQF:** EQF level 7 | **Thesis:** Scheduling for MU-MIMO using hybrid beamforming based on DPC

WORK EXPERIENCE

 *Northeastern University - Institute for the Wireless Internet of Things (WIoT)* – Boston, United States

Visiting Ph.D. Student

[04/23/2025 – 09/30/2025]

Research topics: O-RAN NTN handover optimization, direct-to-satellite communication, THz communications.

 *University of Padova - Department of Physics and Astronomy* – Padova, Italy

University Teaching Assistant

[10/01/2024 – 02/01/2025]

Conducted laboratory sessions for the Machine Learning course

 *University of Padova - Department of Information Engineering* – Padova, Italy

Postgraduate Researcher

[10/01/2021 – 09/30/2022]

Main research topics: Non-Terrestrial networks, 6G wireless networks, mmWave communications.

 *University of Padova - Department of Information Engineering* – Padova, Italy

University Teaching Assistant

[03/01/2022 – 06/01/2022]

Assisted in teaching the Computer Networks course.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING B2 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

SKILLS

Python / Linux OS / Windows OS / C++ / Microsoft Office / Git

PUBLICATIONS

- [1] M. Figaro, F. Rossato, M. Giordani, A. Traspadini, T. Shimizu, C. Mahabal, S. Herath, C. Lee, D. K. Pekcan, and M. Zorzi, "5G NR non-terrestrial networks: From early results to the road ahead," *npj Wireless Technology* (to appear).
- [2] A. Traspadini, M. Zorzi, and M. Giordani, "Performance evaluation of LoRa for IoT applications in non-terrestrial networks via ns-3," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2025 (to appear).
- [3] A. Traspadini, A. A. Deshpande, M. Giordani, C. Mahabal, T. Shimizu, and M. Zorzi, "Sensing-based beamformed resource allocation in standalone millimeter-wave vehicular networks," in *Proceedings IEEE International Conference on Communications (ICC)*, 2025.
- [4] A. Bonora, A. Traspadini, M. Giordani, and M. Zorzi, "Performance evaluation of satellite-based data offloading on Starlink constellations," in *Proceedings IEEE Wireless Communications and Networking Conference (WCNC)*, 2025.
- [5] A. Traspadini, M. Giordani, and M. Zorzi, "Enhanced time division duplexing slot allocation and scheduling in non-terrestrial networks," *58th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Dec. 2024.
- [6] S. Baldoni, F. Battisti, F. Chiariotti, F. Mistrorigo, A. B. Shofi, P. Testolina, A. Traspadini, A. Zanella, and M. Zorzi. Apr. 2024. Questset: A VR dataset for network and quality of experience studies. In *Proceedings of the 15th ACM Multimedia Systems Conference (MMSys '24)*. Association for Computing Machinery, New York, NY, USA.
- [7] A. Traspadini, M. Giordani, G. Giambene, T. De Cola and M. Zorzi, "On the energy consumption of UAV edge computing in non-terrestrial networks," *2023 57th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Dec. 2023.
- [8] A. Traspadini, M. Giordani, G. Giambene and M. Zorzi, "Real-time HAP-assisted vehicular edge computing for rural areas," in *IEEE Wireless Communications Letters*, vol. 12, no. 4, pp. 674-678, April 2023.
- [9] D. Wang, A. Traspadini, M. Giordani, M. -S. Alouini and M. Zorzi, "On the performance of non-terrestrial networks to support the internet of things," *2022 56th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Nov. 2022.
- [10] A. Traspadini, M. Giordani, M. Zorzi "UAV/HAP-assisted vehicular edge computing in 6G: Where and what to offload?" *Joint European Conference on Networks and Communications 6G Summit (EuCNC/6G Summit)*, Jun. 2022.
- [11] P. Testolina, M. Lecci, A. Traspadini, M. Zorzi "An open framework to model diffraction by dynamic blockers in millimeter wave simulations," *IEEE Mediterranean Communication and Computer Networking Conference (MedComNet)*, Jun. 2022.

HONOURS AND AWARDS

[11/26/2024] IEEE Future Networks

Finalist in the Competition on Non-Terrestrial Networks for 6G (NTNYS'24)

[06/09/2025] International Conference on Communications (ICC), IEEE ComSoc

Third place - Four Minute Thesis (4MT) Competition